

SAQQQA

سقا

Trucked water is now the utility for millions of people across MENA. It is still billed on a paper form.

An AI agent that uses the mobile network to check a water delivery really happened, then releases payment for the litres it can prove.

A water delivery is paid for on the word of the person being paid.

A tanker arrives at a camp, a clinic or a hotel. Someone signs a paper form to say it delivered ten cubic metres. That signature is what releases the money.

Nobody measured the water. There is no meter on the truck and no meter on the tank. The number on the form is the number the driver said.

Amal is a WASH officer with 97 sites and one pickup truck. Every Thursday she signs for water she did not see arrive, because that is the process.

This is the state of the art. It is a pen.

UNHCR FORM F-306

GPS coordinates written by hand. A pen signature from a delivery-point representative. Copies posted once a week.

“Will be used as part of the payment justification process to the contractor.”

UNHCR's own words, printed on the form.

One cause, four consequences.

The only record of a delivery is created by the party being paid for it.

WHAT GOES WRONG

WHY NOBODY CATCHES IT

Short-filling	the tanker arrives two thirds full and is paid for a full load. Nobody at the tank has a meter, so there is nothing to disagree with.
Diversion	the load is sold somewhere that pays better and the delivery is logged anyway. Nobody records where the truck actually went.
Ghost trips	trips are billed that never happened at all. The log book is the only record that the trip existed.
A bought signature	the delivery-point representative signs for a few dinars. The signature is the payment trigger, so buying it is the whole attack.

A truck that never turns up is reported within the hour by a thirsty household. These four are the frauds that survive, because nobody who could report them has a reason to.

HOW BIG THIS IS

Drought made the tanker the utility.

\$176M

a year in Jordan's tanker market — more than every public supplier combined

91%

of that water drawn from illegal sources

\$40k

a day for one UNICEF trucking operation in Gaza

4.8–10.7×

the piped price, paid by the people with the least

Iraq has just had its driest year since 1933 and Basra's 4.5 million people are on daily deliveries.

Klassert et al., Nature Sustainability 6:1406 (2023) · UNICEF · independently verified

Measure the water at the tank. Verify the trip with the mobile network.

The payer — an aid agency, a municipality, a hotel — puts two SIM cards into the contract. Neither of them goes on the contractor's vehicle body, and nothing is asked of any household.

SIM ONE, IN THE TRUCK CAB

Lets the operator's network say which truck this was, which well it loaded at, and how long it stood at the tank.

SIM TWO, IN A SENSOR ON THE PAYER'S TANK

A level and turbidity probe. The level rising while the truck is there is the volume delivered, in litres.

An AI agent watches both, decides what it is willing to believe, and releases payment for the litres it can account for. Every decision is written down with its reasons.

What actually happens on a single trip.

01 The truck loads at the contracted well

A Geofencing subscription on the cab SIM fires when it enters and leaves the well zone. If it loaded somewhere else, we know before it arrives.

02 The truck reaches the payer's tank

A second geofence fires. Location Retrieval fills in where it went in between, so a stop that has no business being there is visible.

03 The tank level rises while the truck is standing there

The rise, in litres, is the delivered volume. The dwell time has to be long enough to have moved that much water, or the two facts disagree.

04 The agent checks whether the sensor can be believed at all

Location Verification says it is still where it was installed. Device Reachability separates “offline” from “removed”. Device Swap says its SIM is still in it.

05 The agent checks who is about to be paid

SIM Swap says the payee's number has not been taken over since the contract was signed.

06 Payment is released per litre, with the reasoning stored

Not a score and not a verdict on a person. A number of litres, the evidence for it, and what was paid.

WHAT EACH HALF PROVES

The network does not measure water, and we do not say it does.

HOW MUCH ARRIVED

is a sensor question.

The tank level answers it, in litres. That is a physical measurement and it needs a physical device.

WHOSE TRUCK, FROM WHICH WELL, AND IS THE SENSOR HONEST

are network questions.

Nothing mounted on the truck can answer them honestly, because the truck belongs to the party being audited.

Any claim that the network measures water would be false, and the first engineer in the room would know it.

Our mentor put it plainly: no operator in the region has all the network APIs live. So Saqqa is built to decide on three, and to say what it could not check.

3 APIs

Location Verification, Device Reachability, SIM Swap: the smallest set Saqqa can work on. No geofencing, so dwell is polled every 15 minutes and carried with that uncertainty, never rounded away.

12_{of 12}

trips reach the same verdict as the seven-API build. Every verdict on this profile names the checks it could not run. scripts/compare_profiles.py fails if that ever stops being true.

The screenshot shows the Saqqa web interface. The header includes the Saqqa logo and the tagline 'Network-attested delivery and payment for trucked water'. The main content is divided into two columns. The left column, titled 'Pick a delivery', lists 12 delivery trips with their status (Release, Hold, or Escalate) and a 'start here' link. The right column, titled 'What the hauler claims', shows the 'Payee number SIM-swapped yesterday' and a 'Run verification' button. Below this, it lists 3 CAMARA APIs and their status (Live, In-MENA, Device Swap, Geofencing Subscriptions, Location Retrieval, Location Verification, SIM Swap, Live in MENA). The 'Verdict' section shows a large 'Block' status and a message: 'payee number SIM-swapped inside the 72 h window: identity re-check before any payment'. The 'Next step' section shows a message: 'payment blocked: identity re-check ticket to the payer's finance desk: nothing sent to the payee's number'. The bottom of the page shows 'Not checked on this profile'.

SIM Swap, the one that stops money, is the most deployed CAMARA API in MENA today (GSMA map, 12 Sep). Location Verification is the one that has to land. Four more APIs sharpen the answer: next slide.

The full profile adds four APIs to the core, each answering a question the core cannot.

Location Verification	core · Anti-fraud	presence poll at the tank zone; the sensor is still standing where we installed it
Device Reachability	core · Device intelligence	separates a sensor that is offline from one that has been removed; re-asked before a trace is rejected
SIM Swap	core · Anti-fraud	the payee's number has not been hijacked before the money moves; retrieve-date puts the change on the ticket
Geofencing Subscriptions	full · Device intelligence	entry and exit of the well zone and the tank zone, with dwell. Four per trip; CloudEvents delivered to our webhook
Device Swap	full · Anti-fraud	the sensor's SIM is still inside the sensor, not in someone's handset
Location Retrieval	full · Device intelligence	where the truck was when the tank rose without it. Chosen by the investigation, not run on every trip
Device Roaming Status	full · Device intelligence	a fleet SIM that has left the country entirely

All seven through the official Python SDK, every request and response recorded and shown verbatim. Two of the four CAMARA categories. Both profiles switch with one control on the dashboard.

WHERE THE AI AGENT THINKS

The payment gate is a fixed rule. The investigation is the agent.

A PARTIAL DELIVERY

Driver claimed 10 m³. The tank rose 6. The truck stood there 17 minutes. The contractor has never given us trouble.

Pay for 6, hold the rest, ask the contractor. Do not call it fraud.

AN OPEN CASE

Tank 7 rose 9.7 m³ between 09:05 and 09:27. The operator's stamps put the truck in the zone from 09:35.

Another contractor? A broken sensor? A manipulated trace? A truck nobody registered?

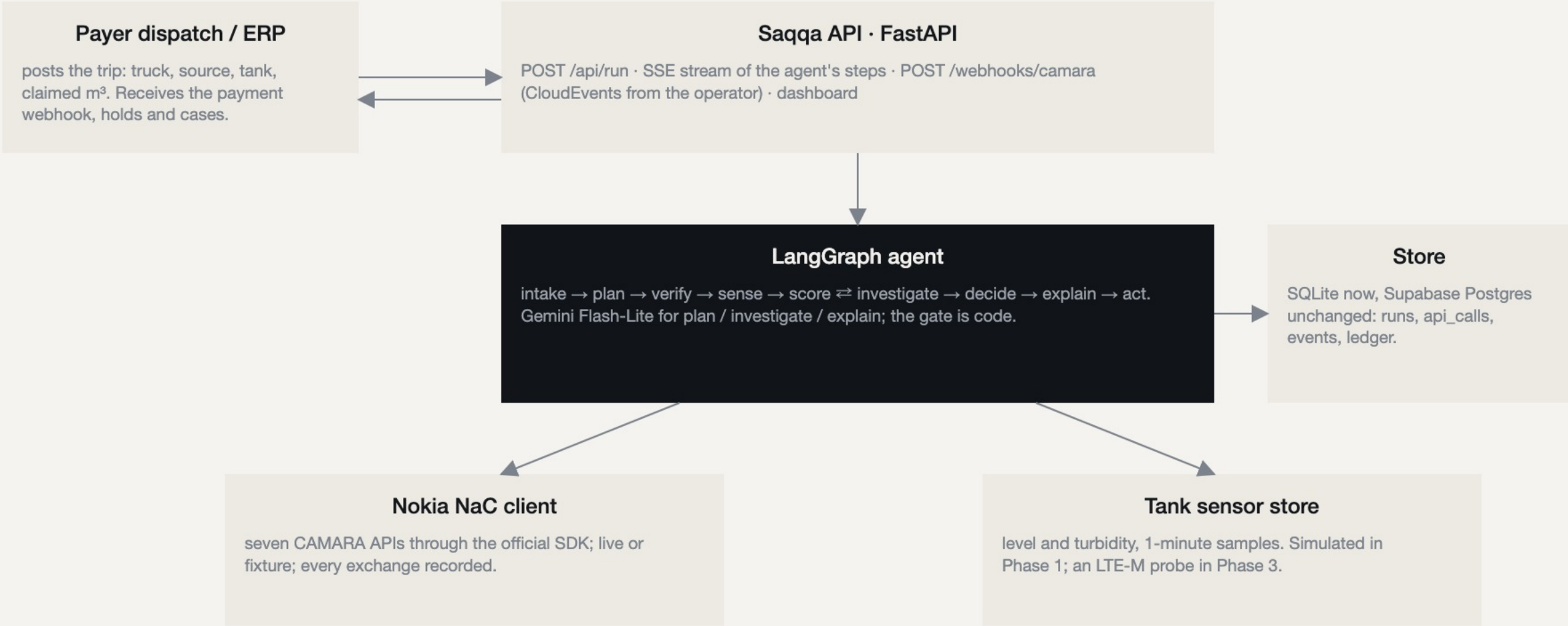
The agent picks which network call to make next, and decides when it has heard enough to stop.

A lookup table cannot pick the next question, and it cannot know when to stop asking. That is the difference between an agent and an if-statement with a model attached.

One graph. The model plans, investigates and explains. Code decides the money.

intake → plan → verify → sense → score ⇔ investigate → decide → explain → act

plan · model	which optional checks to buy for this trip, given site type, contractor history and the cost per call. The mandatory set is fixed.
verify · code	one CAMARA call per tool through the Nokia SDK; request, response, latency and mode recorded; failures returned as evidence, never raised.
score · code	litres, dwell, fill ratio, overlap of the rise with the dwell, pump-time plausibility, capacity accounting, and the contradictions between them.
investigate · model	given the contradictions, the next network question or none; prefers checks that could exonerate an honest contractor. Three rounds at most.
decide · deterministic gate	RELEASE, HOLD, ESCALATE or BLOCK with the litres to pay, in a fixed order of severity an auditor can re-run without a model.
explain and act · model, then code	a two-sentence audit note in English and Arabic; payment webhook, ledger row, F-306-compatible export, supervisor case.



Nokia Network as Code sandbox → operator network: cab SIM, sensor SIM, payee number

no household is ever queried

The run that matters: the agent refuses to believe its own sensor.

Tank 7 rose 9.7 m³ between 09:05 and 09:27. The operator's stamps put the truck in the zone from 09:35. The network wins.
Status line top right: Nokia NaC live, Gemini, webhooks on. Four operator webhooks received for this run's own subscriptions.

Saqqaالسقا | Network-attested delivery and payment for trucked water

Pick a delivery

Twelve real-shaped trips, each with a known outcome. Choose one; it runs when you click it.

01 Honest delivery

Release

→ start here

02 Short fill, 6 of 10

Hold

03 Tank was nearly full

Release

04 Never opened the valve

Escalate

05 Ghost trip

Escalate

06 Substitution: dirty water

Escalate

07 Sensor offline

Hold

08 Sensor SIM moved

Escalate

09 Trip inflation

Escalate

10 Payee SIM-swapped

Block

What the hauler claims

The agent doubts its own sensor

Tank 7 rises 9.7 m³ between 09:05 and 09:27. The operator's timestamps put T-14 in the zone only from 09:35. The measurement is not believed.

trip #4832 · T-14 · claims 10 m³ into tank K7 · Al-Furat Water Transport · camp site

Run verification

core · 3 APIs

full · 7 APIs

trip 10:29

7 CAMARA APIs · dwell stamped by operator geofence events

• Device Reachability Status

• Device Roaming Status live in MENA

• Device Swap

• Geofencing Subscriptions

• Location Retrieval

• Location Verification

• SIM Swap live in MENA

• Live on Nokia Network as Code · LangGraph with gemini-3.5-flash · network events by webhook

Verdict

Escalate

measurement not believed: tank rose 09:05-09:27 while the operator puts the truck in the zone 09:35-10:03; not credited to this truck; possible unregistered delivery or manipulated trace

Next step

case opened for the payer's supervisor with the full trace; contractor notified; no payment

THE DEMO

14

Contradiction found, two questions asked, measurement rejected, nobody accused.

score: “does not fit: tank rose 09:05–09:27 but the truck was in the zone 09:35–10:03” · investigate: reachability re-asked, truck location retrieved · decide: ESCALATE, 0 m³ released

Pick a delivery

Twelve real-shaped trips, each with a known outcome. Choose one; it runs when you click it.

01 Honest delivery

Release

→ start here

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Hold

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09 Trip inflation

Escalate

10 Payee SIM-swapped

Block

→ the one that stops money

11 Agent doubts its sensor

The evidence behind it

What the agent did

Where the truck was

The record

Network calls

12 calls

INVESTIGATE

round 2: ['reachability_status'] · Re-checking sensor reachability could verify if the tank telemetry timing is accurate, potentially exonerating the contractor if the timestamp is off.

VERIFY

reachability_status(+99999991001) → {"reachable": true, "connectivity": ["DATA"]}

SENSE

tank K7 level 38.4 cm at 08:30 · rose 97.4 cm 09:05-09:27 · turbidity max 1.2 NTU · truck dwell 09:35-10:03

SCORE

delta m3 **9.7** claimed m3 **10** fill ratio **0.97** dwell min **28** rise window **09:05 – 09:27** rise dwell overlap min **0** turbidity ntu **1.2**

sensor online **true** sensor in place **TRUE** sensor swapped false payee swapped false capacity ok **true** truck location verify **TRUE**

SCORE

does not fit: tank rose 09:05-09:27 but the truck was in the zone 09:35-10:03

INVESTIGATE

round 3: nothing more the network can tell us; deciding on what we have · The timing contradiction between the tank rise and truck geofence dwell cannot be resolved by network checks as the physical presence and sensor data are already fully verified.

DECIDE

ESCALATE – measurement not believed: tank rose 09:05-09:27 while the operator puts the truck in the zone 09:35-10:03; not credited to this truck; possible unregistered delivery or manipulated trace

EXPLAIN

Decision is ESCALATE because measurement not believed: tank rose 09:05-09:27 while the operator puts the truck in the zone 09:35-10:03; not credited to this truck; possible unregistered delivery or manipulated trace. Measured 9.7 m³ against a 10.0 m³ claim, supported by geofence subscription logs confirming the truck's entry timeline.

ACT

case opened for the payer's supervisor with the full trace; contractor notified; no payment

ACT

ledger written · F-306-compatible export ready · 12 operator calls cited

What it takes to deploy

On the truck

Nothing. The SIM already in the cab is the tracker.

On the tank

One fixed level sensor with its own SIM, bought once by the buyer.

In software

One agent and three CAMARA APIs, running on Nokia Network as

In paperwork

The F-306 form stays; Saqqa fills it from the evidence.

In money

13.6¢ of network calls against a \$30 trip.

TWELVE SEEDED TRIPS, ALL RUN LIVE

Every decision as expected, 115 real CAMARA calls, zero errors.

Honest delivery 9.7 m ³ verified, 31,040 IQD, same day	RELEASE	Sensor offline unverified, not fraud; back to paper	HOLD
Short fill, 6 of 10 pay 6 m ³ now, contractor asked	HOLD	Sensor SIM moved device swapped, not at install point	ESCALATE
Tank was nearly full 5.8 m ³ was all it could take; flag dispatch, not the contractor	RELEASE	Trip inflation 9.8 + 8 m ³ off one 10 m ³ load	ESCALATE
Showed up, never opened the valve 22 min dwell, level flat	ESCALATE	Payee SIM-swapped no money to a hijacked number	BLOCK
Ghost trip fleet SIM never in the zone; roaming abroad	ESCALATE	Trace contradicts the network measurement not believed	ESCALATE
Substitution, dirty water level rose, turbidity 41 NTU	ESCALATE	Manipulated feed 98 cm in one minute	ESCALATE

Unverified is never fraud: an offline sensor sends the trip back to the paper process it came from. The honest hauler is paid the same day; only the ones stealing are bitten.

The sandbox delivers real webhooks. It does not move trucks.

VERIFIED, 2 SEPTEMBER

Every Geofencing subscription the agent creates produced a CloudEvent at our webhook within eight seconds: type area-entered or area-left, the subscription id in the payload, posted from Nokia's platform. Four personas: clean, compromised, UNKNOWN, PARTIAL and unreachable.

SAID ON SCREEN, EVERY RUN

Simulators have fixed positions and fire the subscribed event on creation, so the trip clock is ours and the calls that attest to it are real. The tank trace is a pump model with its parameters on screen; hardware is Phase 3. The location APIs are not yet commercially live with a MENA operator, so Phase 1 runs on the sandbox and says so.

THE OBVIOUS OBJECTION

“Why not just bolt a GPS tracker on the truck?”

BECAUSE RESOLUTION IS THE WRONG AXIS.

A tracker reports its own position. Anything that reports its own position can be made to lie, and a tamper-evident seal stops a screwdriver, not a spoofed signal.

2,000 metres you cannot forge beats 5 metres you report yourself, when you are the party being audited.

Cell-level location is coarse, roughly two kilometres. It is computed by infrastructure the contractor does not own and cannot reach. And the hardware sits on the payer's tank, so the truck carries nothing to disable, swap or drive around with.

AGAINST THE ALTERNATIVES

Nothing else answers both halves at once.

	PROVES LITRES	PROVES WHOSE TRUCK	CONTRACTOR CAN FAKE IT	COST PER SITE
Paper log (F-306)	claimed only	signature	yes, trivially	near zero
Fleet GPS tracker	no	yes, to metres	yes: the device reports its own position	per truck, ongoing
Biometric / RFID gate	no	at a fixed gate only	no, but tanks are not gated	per gate
Saqqa	yes, measured	yes, at zone scale	no: the operator computes it	one sensor, payer-owned

Only the last row answers both questions at once, and it is the only row where the party being paid does not produce the evidence.

Every component of the prototype, and what it does.

LangGraph	the agent: one stateful graph, conditional edges for the investigate loop, the CAMARA calls as its tools
Gemini 3.5 Flash-Lite · Flash · Groq Llama 3.3 70B	planning, investigation choice, audit notes. In that fallback order, per call; Gemini 2.5 Flash is no longer offered to new accounts
Nokia NaC Python SDK 10.0	the seven CAMARA APIs, live against the sandbox; responses normalised and recorded verbatim
FastAPI · SQLite → Supabase	REST, Server-Sent Events for the live trace, the CloudEvents webhook receiver; runs, calls, events and the ledger
Leaflet · the dashboard	operator geofences on a map, the tank trace with the dwell window, the reasoning as it streams, every call, the decision, F-306 export
Docker · Render · cloudflared	one-command run, one-click deploy, a public sink for the operator's webhooks during development

The payer already exists, and already pays against paper.

WHO THE BUYER IS

75%

of Jordan's tanker sales serve businesses: hotels, factories, sites. They pay per cubic metre and get short-filled.

21%

funded: the humanitarian buyer whose donors demand third-party proof of every dollar.

\$30–70

a delivery today, paid against a signature.

New

API revenue for operators. UN agencies and utilities are a customer they do not have today.

HOW WE CHARGE

Per verified delivery

cents against a \$30–70 trip; verification costs about 13 cents. The price scales with value delivered rather than with seats.

Platform fee per instrumented tank

a flat monthly fee per tank on the payer's own site, which is where the sensor lives. One sensor covers every contractor who delivers to it.

Operator revenue share

the operator bills the CAMARA calls and keeps a share. That is what makes us worth onboarding rather than merely tolerating.

One sensor per tank, not per contractor and not per delivery.

WHY THE UNIT ECONOMICS IMPROVE WITH VOLUME

The sensor is fixed to the payer's tank, so it is bought once and then covers every contractor who ever delivers to that tank. Cost per verified delivery falls as the tank gets busier. A camp taking two loads a day amortises the same hardware twenty times faster than one taking two a week.

WHY IT SPREADS INSTEAD OF STALLING

Nothing to install on the fleet

the contractor's only obligation is a SIM in the cab. No hardware to buy, fit, maintain or argue about, which is what kills tracker rollouts at procurement.

The operator is the channel

once one operator exposes these APIs in a market, every payer in that market is reachable with no new integration from us. We sell through the operator, not around it.

Same engine, other cargo

trucked diesel and LPG ask the same question with a bigger number: a 20,000-litre load is worth about \$15,000, generator tanks already carry level sensors, and the same 13 cents of calls verifies it. Food aid and cold chain follow.

Sequenced on GSMA's deployment database, not on our roadmap: Iraq first, then Jordan as the commercial market, then the Gulf.

Saqqqa is not surveillance of haulers.

It is same-day settlement for the honest ones.

Today an honest driver is paid on a weekly paper cycle he cannot influence, and when a tank comes up short he has no way to prove it was full when he left.

Under Saqqqa he is paid the same day, and he holds a record the operator stands behind that says he delivered. It only bites the ones who were stealing.

Nothing new on the truck, nothing new in the law.

On the truck: nothing	the SIM already in the cab is the tracker. No hardware to buy, fit, maintain or argue about at procurement.
On the tank: one sensor	a fixed level sensor with its own SIM on the buyer's site, bought once, covering every contractor who ever delivers there.
In software: one agent, three APIs	runs on Nokia Network as Code today; every verdict is reached on the three-API core profile. A pilot needs one buyer, one hauler and one operator API key.
In paperwork: nothing changes	the F-306 delivery form stays the legal record; Saqqa fills it from the evidence instead of from a signature.
In money: 13.6 cents a trip	measured, against a \$30 delivery: under half a percent, paid per verified trip, with the operator earning per call.

TEAM STARX

Four engineering students, Indian Institute of Technology Delhi,
Abu Dhabi.

James Joshua Koshy

Computer Science

NETWORK INTEGRATION AND AGENT

Works out what the network can actually tell us, wires the CAMARA APIs into the agent, and ran the live verification.

Evan Johan Tobias

Computer Science

AGENT AND BACKEND

Builds the agent, the rule that decides what gets paid, and the records behind it.

Divyam Thakur

Computer Science

RESEARCH AND SOURCING

Chases down every number in this deck and where it came from, including the ones that argue against us.

Kirti Roshankumar Thakar

Chemical Engineering

DESIGN AND PITCH

Designs the deck and decides how the pitch gets told.